Case Study: Chronic air pollution exposure in relation to mortality in the Nurses' Health Study: survival analysis with mis-measured covariate histories

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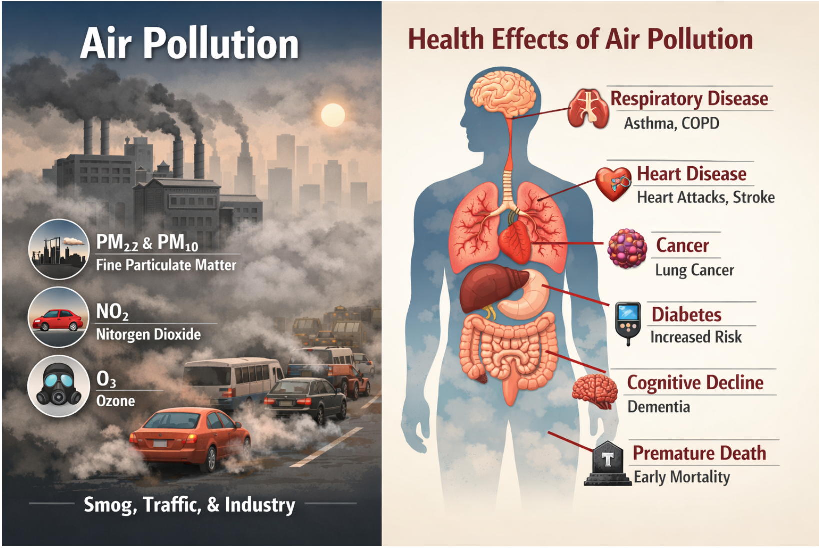
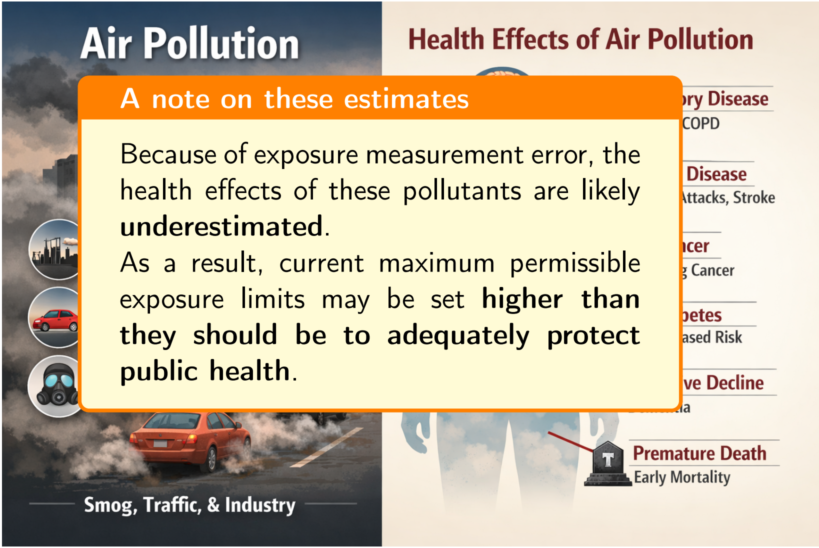
Air Pollution and Health



Air Pollution and Health



A note on these estimates

Because of exposure measurement error, the health effects of these pollutants are likely **underestimated**.

As a result, current maximum permissible exposure limits may be set **higher than they should be to adequately protect public health**.

Background: Environmental Epidemiology

Environmental epidemiologists are interested in estimating the effect of time-varying exposure to air pollution constituents in relation to chronic diseases including lung cancer, cardiovascular disease incidence and all-cause mortality.

Nurses' Health Study has been an important source of information about the health effects of air pollution, e.g.

Robin C. Puett, Joel Schwartz, Jaime E. Hart, Jeff D. Yanosky, Frank E. Speizer, Helen Suh, Christopher J. Paciorek, Lucas M. Neas, Francine Laden, Chronic Particulate Exposure, Mortality, and Coronary Heart Disease in the Nurses' Health Study, American Journal of Epidemiology, Volume 168, Issue 10, 15 November 2008, Pages 1161–1168

Puett RC, Hart JE, Yanosky JD, Spiegelman D, Wang M, Fisher JA, Hong B, Laden F. Particulate matter air pollution exposure, distance to road, and incident lung cancer in the nurses' health study cohort. Environ Health Perspect. 2014 Sep;122(9):926-32.

Main Study: Nurses' Health Study

The Nurses' Health Study (NHS) began in 1976 with 121,700 female registered nurses who lived in 11 states (CA, TX, FL, MA, PA, OH, NY, NJ, MI, CT, MD).

- Exposure of interest, $x(t)$, is 12 month moving average of *personal* ambient $PM_{2.5}$ exposure
- The surrogate exposure, $X(t)$, is a time-varying function of monthly average $PM_{2.5}$ concentration, $C(t)$, from the nearest EPA monitor or from the Yanosky model.
- $PM_{2.5}$ from nearest EPA monitors measures ambient personal $PM_{2.5}$ exposures with error, $X(t) \neq x(t)$

Yanosky, J.D., Paciorek, C.J., Suh, H.H. (2009). Predicting chronic fine and coarse particulate exposures using spatio-temporal models for the Northeastern and Midwestern United States. Environmental Health Perspectives, 117(4), 522–529.

External air pollution exposure validation study

First validation data set consisted of data from 9 cities: Baltimore,MD; Boston,MA; Steubenville,OH; Atlanta,GA; Los Angeles,CA; Seattle,WA; RTP,NC; Elizabeth, NJ; Houston, TX

Kioumourtzoglou MA, Spiegelman D, Szpiro AA, Sheppard L, Kaufman JD, Yanosky JD, Williams R, Laden F, Hong B, Suh H. Exposure measurement error in PM2.5 health effects studies: a pooled analysis of eight personal exposure validation studies. Environ Health. 2014 Jan 13;13(1):2;

Recently expanded to include data from 17 communities

Zhang B, Eum K. D, Szpiro A A, Zhang N, Hernández-Ramírez RU, Spiegelman D, Suh H. 04/07/2025. Exposure measurement error in air pollution health effect studies: a pooled analysis of personal exposure validation studies in 17 communities across the United States. International Journal of Environmental Health Research, 1–11.

Table 1: Validation studies used in our analyses

Cities	Years	# Subjects	Sample Session Duration	Age Mean (S.D.)
Atlanta, GA	1999–2000	31	7 days	65.0 (13.5)
Baltimore, MD	1998–1999	35	12 days	70.8 (7.7)
Boston, MA	1999–2000	56	7 or 12 days	62.3 (14.1)
Los Angeles, CA	2000–2002	37	7 days	56.3 (13.9)
RTP, NC	2000–2001	37	7 days	64.5 (7.8)
RIOPA	1999-2001		48 hr	
Los Angeles, CA		73		46.1 (18.6)
Elizabeth, NJ		57		48.2 (17.8)
Houston, TX		62		48.5 (16.6)
Seattle, WA	2000–2001	89	10 days	76.7 (6.5)
Steubenville, OH	2000	28	48 hr/we for 12 we	71.0 (10.0)

Exposure definition for the external air pollution exposure validation study

- The true exposure, $c(t)$, is the 24-hour integrated *point* personal $PM_{2.5}$ concentrations.
- The surrogate exposure, $C(t)$, is the average monthly $PM_{2.5}$ concentration from nearest EPA monitor to the residence at the same calendar time.

Methods presented in this talk

- Survival data analysis in prospective cohort studies
- Time-varying functions of the exposure history are the exposures of interest, for example the 12 month moving average :

$$x(t) = \sum_{s=t-11}^{t} \frac{c(s) \times I(s)}{\sum_{s=t-11}^{t} I(s)},$$

where $I(s) = 1$ if c(s) is measured at time s, 0 o.w.

- The exposure variable is measured with error

Main study/external validation study design

In the main study:

$$\{C_i(t), W_i(t), T_i, D_i\},\ i = 1, \cdots, n_1\ ;\ t = t_{i0}, t_{i1}, \cdots, T_i,$$

where t_{i0} is the age at entry into the cohort, the time scale is, here, age in months (Korn et al., 1997), and $X_i(t)$ is a function of $\{C_i(s), s \leq t\}$.

- $W(t)$: error-free covariates,
- D: the indicator for failure,
- T_i^0: failure time, $\mathcal{C}_i$: censoring time, t^*: the end of follow-up.
- T_i: follow-up time, where $T_i = \min(T_i^0, \mathcal{C}_i, t^*)$,

In the external validation study:

$$\{c_i(t), C_i(t), W_i(t), t_i\},\ i = n_1 + 1, \cdots, n_1 + n_2;\ t = t_{i0}, t_{i1}, \cdots, T_i),$$

Notation

Exposure	True	Surrogate
Point	$c(t)$	$C(t)$
12 month moving average	$x(t)$	$X(t)$

In general, $x(t) = f[c(t)]$, $X(t) = f[C(t)]$, and $f(\cdot)$ is some function of the exposure history.

The 12 month moving average is

$$x(t) = \int_{max(t_0,t-11)}^{t} c(s)\,ds \text{ and } X(t) = \int_{max(t_0,t-11)}^{t} C(s)\,ds$$

and the discrete versions are

$$x(t) = \sum_{max(t_0,t-11)}^{t} \frac{c(s) \times I(s)}{\sum_{t-11}^{t} I(s)} \text{ and } X(t) = \sum_{max(t_0,t-11)}^{t} \frac{C(s) \times I(s)}{\sum_{t-11}^{t} I(s)},$$

Methods

Ordinary regression calibration(ORC)

- First proposed by Prentice (1982, *Biometrika*), more development followed by Spiegelman et al., Wang et al.
  1. Spiegelman D, McDermott A, Rosner B (1997), Regression calibration method for correcting measurement error bias in nutritional epidemiology *Am J Clin Nutr* 65(suppl):1179S-1186S.
  2. Wang CY, Hsu L, Feng ZD, Prentice RL (1997) Regression calibration in failure time regression, *Biometrics*, 53(1), 131-45.
- Valid for measurement error correction with time-independent exposures in main study / external validation study designs.
- Not applicable for use with mis-measured time-varying exposures.
- Software : SAS macros `%blinplus` and `%relibpls`. (`http://ysph.yale.edu/cmips/research/software`)

Methods

Risk set regression calibration(RRC)

- Re-calibrates the measurement error model within each risk set.
- Xie SX, Wang CY and Prentice RL (2001, JRSSB) developed this for the classical additive measurement error model, $C = c + e$, for the main study / internal reliability study design, for time-independent exposures.
- Liao XM, Zucker MD, Li Y and Spiegelman D (2011, Biometrics) extended this to time-varying point exposures in the main study/external validation study design under a more general measurement error model.
- We have further extended the risk set regression calibration method to correct for measurement error in time-varying functions of the exposure history, tackling the limitations of the data available in current validation studies.

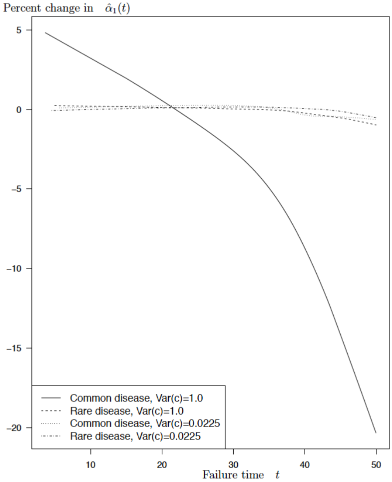
Basic idea

The measurement error model evolves over time

- Even for a baseline exposure that is associated with the risk for outcome, measurement error in the cohort evolves over time
  - Error among those with truly high exposure increases
  - Error among those with truly low exposure decreases

Ref:
1. *Xie SX, Wang CY and Prentice RL 2001,63(4), 855-870, Journal of the Royal Statistical Society: Series B.*
2. *Liao XM, Zucker MD, Li Y and Spiegelman D, 2011, 67, 50-58, Biometrics.*

Risk set Regression Calibration



Change in the deattenuation factor over time, in the model

$\mathrm{E}(c(t)|C(t)) = \alpha_0(t) + \alpha_1(t)C(t)$ (simulated data)

Main challenge

Basic idea

In a prospective study, the measurement error model must be refit at each time a case occurs, because the working measurement error model after baseline evolves as a function of who is still at risk.

- It is not possible to validate exposure metrics that are time-varying functions of the exposure history. Few exposures have been repeatedly validated in the same study participants. e.g. $X_i(t_k) = \sum_{s=t_k-11}^{t_k} \frac{C(s)\times I(s)}{\sum_{s=t_k-11}^{t_k} I(s)}$,

  where $t_0 =$ age at entry into the study, $t_k =$ current age.

- Challenge: Estimating $\hat{x}(t)$ in the main study using information that is available in the validation study, so β can be estimated without measurement error bias from $\lambda(t; \hat{x}(t), \boldsymbol{W}(t))$, in the model

$$\lambda(t; x(t)) = \lambda_0(t) \exp\{\beta x(t)\}, \quad t \geq 0.$$

Our solution

Illustrating the idea with 12 month moving average exposure:

$$\begin{aligned}
& \lambda(t_k|C(t_{k-11}), C(t_{k-10}), \cdots, C(t_k)) \\
= \quad & E\left[\lambda(t_k|c(t_{k-11}), c(t_{k-10}), \cdots, c(t_k))\middle|\mathrm{C}_k, T \geq t_k\right] \quad \text{(due to surrogacy assumption)} \\
= \quad & \lambda_0(t_k) E\left[\exp\left(\beta \cdot \frac{1}{12}\sum_{s=0}^{11} c(t_{k-s})\right)\middle|\mathrm{C}_k, T \geq t_k\right] \\
\approx \quad & \lambda_0(t_k) \exp\left(\beta \cdot \frac{1}{12}\sum_{s=0}^{11} E\left[c(t_{k-s})\middle|\mathrm{C}_k, T \geq t_k\right]\right) \\
= \quad & \lambda_0(t_k) \exp(\boldsymbol{\beta}\hat{x}(t_k))
\end{aligned}$$

where $\mathrm{C}_k = (C(t_{k-11}), C(t_{k-10}), \cdots, C(t_k))$.

The last approximation holds when one of the following four conditions holds in each risk set: (1) normally distributed errors for $\mathrm{c}_k|\mathrm{C}_k$ with constant variance; (2) normally distributed errors for $\mathrm{c}_k|\mathrm{C}_k$ with small variance; (3) Small RR and $\mathrm{c}_k|\mathrm{C}_k$ has small variance; (4) Small RR and $\mathrm{c}_k|\mathrm{C}_k$ has constant variance.

Our solution

If a linear measurement error model for the point exposures fits the data :

$$\mathrm{E}(c(t)|C(t)) = \alpha_0(t) + \alpha_1(t)C(t),$$

then it implies the following localized error assumption in each risk set :

$$\begin{aligned}\mathrm{E}(c(t_j)|C(t_{k-11}), \cdots, C(t_k), T \geq t_k) &= \mathrm{E}(c(t_j)|C(t_j), T \geq t_k) \\ &= \alpha_0(t_j) + \alpha_1(t_j)C(t_j) \quad \text{for} \quad k-11 \leq j \leq k.\end{aligned}$$

Hence, $\hat{\mathrm{c}}_k = \mathrm{E}(\mathrm{c}_k|\mathrm{C}_k, T \geq t_k) = \left[\begin{array}{c} \hat{\alpha}_0(t_{k-11}) + \hat{\alpha}_1(t_{k-11})C(t_{k-11}) \\ \hat{\alpha}_0(t_{k-10}) + \hat{\alpha}_1(t_{k-10})C(t_{k-10}) \\ \cdots \\ \hat{\alpha}_0(t_k) + \hat{\alpha}_1(t_k)C(t_k) \end{array}\right]$ and it follows that $\hat{x}(t_k) = \hat{\mathrm{c}}_k' 1_k/12$. We then fit the model

$$\lambda(t_k|C(t_{k-11}), C(t_{k-10}), \cdots, C(t_k)) = \tilde{\lambda}_0(t_k)\exp(\beta\hat{x}(t_k)).$$

Sandwich Variance Estimator

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}_{rrc}) = \frac{1}{n_1}\hat{\mathrm{I}}_{\boldsymbol{\beta}}^{-1}\hat{\mathrm{H}}_{\boldsymbol{\beta},\boldsymbol{\alpha}}\hat{\mathrm{I}}_{\boldsymbol{\beta}}^{-1}$$

Components

$$\hat{\mathrm{I}}_{\boldsymbol{\beta}}^{-1} = \left[\frac{1}{n_1}\sum_{i=1}^{n_1}\frac{\partial \mathrm{U}_{\boldsymbol{\beta}i}(\boldsymbol{\beta}|\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}}\right]^{-1}_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}$$

$$\hat{\mathrm{H}}_{\boldsymbol{\beta},\boldsymbol{\alpha}} = \hat{\mathrm{H}}_{\boldsymbol{\beta}} + \frac{1}{n_1 n_2}\hat{\mathrm{U}}^*(\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\alpha}})\hat{\mathrm{V}}_{\boldsymbol{\alpha}}\hat{\mathrm{U}}^{*T}(\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\alpha}})$$

$$\hat{\mathrm{H}}_{\boldsymbol{\beta}} = \frac{1}{n_1}\sum_{i=1}^{n_1}\tilde{\mathrm{U}}_{\boldsymbol{\beta}i}\tilde{\mathrm{U}}_{\boldsymbol{\beta}i}^{T}$$

$$\hat{\mathrm{U}}^*(\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\alpha}}) = \sum_{i=1}^{n_1}\frac{\partial \mathrm{U}_{\boldsymbol{\beta}i}(\boldsymbol{\beta}|\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}}\bigg|_{(\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\alpha}})}$$

Adjusted score

$$\tilde{\mathrm{U}}_{\boldsymbol{\beta}i} = \mathrm{U}_{\boldsymbol{\beta}i} - \sum_{j=1}^{n_1}\frac{D_j Y_m(i,T_j)\exp(\boldsymbol{\beta}_1'\hat{\mathrm{x}}_i(T_j))}{n_1 S^{(0)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_j)}\left(\hat{\mathrm{x}}_i(\hat{\boldsymbol{\alpha}},T_j) - \frac{\mathrm{S}^{(1)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_j)}{S^{(0)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_j)}\right)$$

Bias-corrected estimator of Mancl & DeRouen (2001) is used for $\hat{\mathrm{V}}_{\boldsymbol{\alpha}}$. (Lin & Wei 1989; Zucker & Spiegelman 2008)

Analysis: Selection of risk set size, s, in the validation study

- After deleting person-months corresponding to ages less than the youngest death, since the validation study was small ($n_v = 100$ participants with 184 person-month observations), we grouped the risk sets to stabilize results, to ensure that each risk set would have a minimum of s observations.
- Different choice of s, the risk set size, results in different compositions and numbers of risk sets in the validation study.
- To determine the optimal value of s, we conducted a 20-fold cross-validation, and calculated the mean square error, for each choice of s ranging from 10 to 100.
- The mean square error was the smallest when $s = 85$, resulting in 2 risk sets, where the first risk set had 86 person months with age ranging from 53.8 to 70.5 years, and the second had 98 person months with age ranging from 70.6 to 87.0 years.
- We proceeded with the analysis with this optimized s.

Basic Characteristics: Main study

Chronic particulate exposure of ambient origin and all-cause mortality in the Nurses' Health Study (2000-2006) and external validation study

Person-months ($n_1 = 108,765$)	Duration of Follow-up	# of deaths	Mean(s.d.) ($\mu g/m^3$)	Range ($\mu g/m^3$)
7,538,537	2000-2006	8604	12.7(3.1)	(0.3,88.0)

Basic Characteristics of Nurses' Health Study (2000.6–2006.5)

Demographics
Age (years): 69.0 (7.3)
White: 94%
BMI (kg/m^2) : 26.8(5.4)

Air Pollution Exposure
Model $PM_{2.5}$ ($\mu g/m^3$): 12.0 (2.8)
Nearest EPA $PM_{2.5}$ $\mu g/m^3$: 12.7 (3.1)

Smoking
Never: 44%
Current: 10%
Former: 45%
Pack-years (ever smokers): 24.9 (21.7)

Health Conditions
Hypercholesterolemia: 61%
Diabetes: 11%
Hypertension: 53%
Family history of MI: 33%

Physical Activity (MET hr/week)
<3 : 24%
3–$<$9: 22%
9–$<$18: 20%
18–$<$27: 12%
$\geq$27: 21%

Diet & SES
Alternative healthy eating index: 195 (78)
Median family income ($1000): 63(24)
Median house value ($10,000): 17(13)

Family Background
Mom housewife: 64%
Dad professional/manager: 26%

Education & Marriage
RN degree: 94%
Married: 82%

Husband Education
$<$HS: 6%
HS graduate: 40%
$>$HS: 54%

Nine City validation study of personal $PM_{2.5}$ exposure in relation to nearest monitor

Table 3: Season-adjusted calibration factors for personal $PM_{2.5}$ of ambient origin

	Monitor $PM_{2.5}$	Model Predicted $PM_{2.5}$
Personal $PM_{2.5}$ of ambient origin	5 cities (255 person-months)	
Estimate (95% CI)	0.31 (0.14, 0.47)**	0.54 (0.42, 0.65)**
p-value for between-city heterogeneity	0.0034	0.1114
$R_I^{\dagger}$	10.75%	0.96%
Total Personal $PM_{2.5}$	9 cities (913 person-months)	
Estimate (95% CI)	0.56 (0.24, 0.88)**	0.81 (0.49, 1.12)
p-value for between-city heterogeneity	0.0084	0.1712
$R_I^{\dagger}$	2.54%	1.28%

* $p\text{-value}_1 < 0.05$, ** $p\text{-value}_1 < 0.01$ for significant difference from 1

† R_I: the proportion of variance explained by the between-cities heterogeneity

The SAS macro %RRC

https://ysph.yale.edu/cmips/research/software/measurement-error-and-misclassification-correction/rrc/

```
  %macro rrc
(
id =,            /* Name of subjects' id */
main =,          /* Name of main data set, required */
validation =,    /* Name of validation data set, required */
surrogate =,     /* exposure with error in main and validation study, required */
true =,     /* exposure without error in validation study, required */
confounder=, /* perfectly measured covariates adjusted in the model, optional */
case =,   /* variable that indicates whether an event occurs or not, required */
time =,   /* variable for time-to-event outcome, required */
type =,    /* types of exposure interested:
         1: Cumulatively average updated exposure (time-varying);
         2: a-unit moving average exposure (time-varying);
         3: Cumulatively total updated exposure (time-varying);
         4: Simple updated exposure (time-varying);
         5: Baseline exposure (time-independent); required */
month=12, /* Required only for type=2. The value of unit a in a-unit moving average (the defau
             Time unit could be month or year or else.  */
missing=,  /* Required only for type=2. Missing indicator for main exposure,
             it's 1 if missing, otherwise it's 0.*/
period=,   /* Required only for type=2. Period indicator for a-unit moving average.
             This variable contains values of -a+1,-a+2,0,1,2,a,a+1,a+2,. */
residual=,  /* if residual=1, use residual method to control for confounding;
             otherwise, residual=0. Required. */
groupnum= 5,  /* the minimum number of subjects in each validation study risk set, optional,
\default=5 */
increments=1, /* the units in which the RR for true exposure will be reported, optional,
\default=1 */
filename=RRCOutput.txt   /* the name of the output file */
);
```

Results

★ Output – Fully adjusted model:

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
This analysis is for 1 - 12 month average exposure
with general linear measurement error model.

The increment for each exposure is: 10.0

There is one exposure and one or more covariates in the analysis.
The exposure is measured with error, but the covariates are error-free.

# of subjects in the main study: 108767
# of person-time observations in the main study: 7538226
# of cases in the main study: 8617
# of unique failure time: 367
```

Results

```
The unique failure times are:
 646.0 651.0 652.0 653.0 655.0 659.0 660.0 663.0 664.0 665.0
 666.0 667.0 669.0 670.0 671.0 672.0 673.0 674.0 675.0 676.0
 677.0 678.0 680.0 681.0 682.0 683.0 684.0 685.0 686.0 687.0

        ......              ......              ......

 989.0 990.0 991.0 992.0 993.0 994.0 995.0 996.0 997.0 998.0
 999.0 1000.0 1001.0 1002.0 1003.0 1004.0 1005.0 1006.0 1007.0 1008.0
 1009.0 1010.0 1011.0 1012.0 1013.0 1014.0 1015.0 1016.0 1017.0 1018.0
 1019.0 1020.0 1021.0 1022.0 1023.0 1024.0 1044.0

 # of subjects in the validation study: 118
 # of person-time observations in the validation study: 215
 # of risk sets in the validation study: 2

 -------------------------------
 Information about the measurement error model
 fitting in the validation study:
 -------------------------------

Risk_set  Age_low  Age_up  Person_year  Intercept  slope     R^2
   1       646.0   846.0        86        0.352    0.356    0.532
   2       847.0  1044.0        98        0.240    0.505    0.448
```

Results

	Naive Cox			RRC		
Variable	β(SE)	HR(95% CI)	p-value	β(SE)	HR(95% CI)	p-value
fine_jeff	**0.125(0.039)**	**1.133(1.05,1.22)**	**0.002**	**0.165(0.073)**	**1.179(1.02,1.36)**	**0.02**
reg2	−0.004(0.032)	0.996(0.94,1.06)	0.90	−0.007(0.036)	0.993(0.92,1.07)	0.84
reg3	0.078(0.035)	1.081(1.01,1.16)	0.03	0.093(0.039)	1.098(1.02,1.18)	0.02
reg4	0.034(0.029)	1.035(0.98,1.10)	0.24	0.022(0.031)	1.022(0.96,1.09)	0.48
heatsn	0.071(0.022)	1.074(1.03,1.12)	0.001	0.072(0.022)	1.074(1.03,1.12)	0.001
yr	0.005(0.007)	1.005(0.99,1.02)	0.42	0.005(0.007)	1.005(0.99,1.02)	0.44
white	0.076(0.045)	1.079(0.99,1.18)	0.09	0.074(0.045)	1.077(0.99,1.18)	0.10
mifamhis	0.115(0.023)	1.122(1.07,1.17)	<0.001	0.115(0.023)	1.122(1.07,1.17)	<0.001
chol	−0.119(0.023)	0.888(0.85,0.93)	<0.001	−0.119(0.023)	0.888(0.85,0.93)	<0.001
conbmi	−0.029(0.002)	0.971(0.97,0.98)	<0.001	−0.029(0.003)	0.971(0.97,0.98)	<0.001
misbmi	−1.214(0.091)	0.297(0.25,0.36)	<0.001	−1.213(0.100)	0.297(0.24,0.36)	<0.001
smkcat2	0.089(0.030)	1.093(1.03,1.16)	0.003	0.088(0.029)	1.092(1.03,1.16)	0.002
smkcat3	0.034(0.046)	1.035(0.95,1.13)	0.46	0.034(0.046)	1.035(0.95,1.13)	0.46
smkcatm	0.127(0.234)	1.136(0.72,1.80)	0.59	0.125(0.234)	1.133(0.72,1.79)	0.59
pkyr	0.014(0.001)	1.014(1.01,1.01)	<0.001	0.014(0.001)	1.014(1.01,1.01)	<0.001
mispkyr	0.240(0.082)	1.272(1.08,1.49)	0.003	0.241(0.082)	1.272(1.08,1.49)	0.003

Results (continued)

	Naive Cox			**RRC**		
Variable	β(SE)	HR(95% CI)	p-value	β(SE)	HR(95% CI)	p-value
ahei	−0.003(0.000)	0.997(1.00,1.00)	<0.001	−0.003(0.000)	0.997(1.00,1.00)	<0.001
misahei	−0.030(0.052)	0.971(0.88,1.07)	0.57	−0.030(0.052)	0.971(0.88,1.08)	0.57
phyact2	−0.492(0.030)	0.611(0.58,0.65)	<0.001	−0.492(0.030)	0.611(0.58,0.65)	<0.001
phyact3	−0.723(0.035)	0.485(0.45,0.52)	<0.001	−0.724(0.035)	0.485(0.45,0.52)	<0.001
phyact4	−0.919(0.048)	0.399(0.36,0.44)	<0.001	−0.920(0.048)	0.399(0.36,0.44)	<0.001
phyact5	−0.910(0.040)	0.402(0.37,0.43)	<0.001	−0.911(0.040)	0.402(0.37,0.43)	<0.001
phyactm	−0.373(0.059)	0.689(0.61,0.77)	<0.001	−0.373(0.059)	0.688(0.61,0.77)	<0.001
hsv2	−0.040(0.037)	0.961(0.89,1.03)	0.27	−0.041(0.037)	0.960(0.89,1.03)	0.26
hsv3	−0.062(0.042)	0.940(0.86,1.02)	0.14	−0.064(0.042)	0.938(0.86,1.02)	0.13
hsv4	−0.001(0.047)	0.999(0.91,1.10)	0.98	−0.004(0.047)	0.997(0.91,1.09)	0.94
hsv5	−0.098(0.057)	0.907(0.81,1.01)	0.09	−0.100(0.057)	0.905(0.81,1.01)	0.08
hsvm	0.194(0.232)	1.214(0.77,1.91)	0.40	0.202(0.232)	1.223(0.78,1.93)	0.38
inc2	−0.011(0.035)	0.989(0.92,1.06)	0.76	−0.010(0.035)	0.990(0.92,1.06)	0.77
inc3	0.007(0.041)	1.007(0.93,1.09)	0.86	0.009(0.041)	1.009(0.93,1.09)	0.83
inc4	−0.078(0.046)	0.925(0.84,1.01)	0.09	−0.076(0.046)	0.927(0.85,1.01)	0.10
inc5	−0.085(0.054)	0.919(0.83,1.02)	0.12	−0.083(0.053)	0.921(0.83,1.02)	0.12
momw2	0.039(0.023)	1.040(0.99,1.09)	0.09	0.039(0.023)	1.040(0.99,1.09)	0.09
dadw2	0.025(0.026)	1.026(0.97,1.08)	0.33	0.026(0.026)	1.026(0.98,1.08)	0.32
educm	−0.229(0.187)	0.795(0.55,1.15)	0.22	−0.232(0.194)	0.793(0.54,1.16)	0.23
rn	−0.079(0.061)	0.924(0.82,1.04)	0.20	−0.079(0.061)	0.924(0.82,1.04)	0.20
mstatm	−0.043(0.182)	0.958(0.67,1.37)	0.81	−0.041(0.188)	0.960(0.66,1.39)	0.83
marry	−0.151(0.030)	0.860(0.81,0.91)	<0.001	−0.152(0.030)	0.859(0.81,0.91)	<0.001
hused2	−0.159(0.048)	0.853(0.78,0.94)	0.001	−0.159(0.048)	0.853(0.78,0.94)	0.001
hused3	−0.199(0.048)	0.819(0.75,0.90)	<0.001	−0.200(0.048)	0.819(0.75,0.90)	<0.001
husedm	−0.205(0.055)	0.815(0.73,0.91)	<0.001	−0.205(0.055)	0.815(0.73,0.91)	<0.001

Summary of Results

NHS 2000–2006, 12-month moving average $PM_{2.5}$ of ambient origin in relation to all-cause mortality

$n/N = 8617$ deaths / 108,767 participants during 7,538,226 person-months

	Age-adjusted model			Fully adjusted model		
	RR	95% CI	p	RR	95% CI	p
Uncorrected	1.20	[1.11,1.29]	< 0.001	1.13	[1.05,1.22]	0.002
Corrected	1.27	[1.08,1.48]	0.003	1.18	[1.02,1.36]	0.024

Age-adjusted model: age (months), calendar year, region, season.
Fully adjusted model additionally includes smoking status, pack-years, family history of MI, BMI, hypercholesterolemia, median family income, median house value, physical activity, race, AHEI, and individual-level SES.

Summary

- Methods are available for measurement error correction of diet, activity, and air pollution effects for time-varying functions of the the exposure history estimated in prospective cohort studies using survival data analysis methods, particularly for the main study/external validation study design.
- We have developed a SAS macro `%rrc` for the method and it can handle four types of exposure metrics:
  - Cumulatively average updated exposure (time-varying)
  - Cumulatively total updated exposure (time-varying)
  - Δ-month moving average exposure (time-varying)
  - Simple updated exposure (time-varying)
  - Baseline exposure (time-independent)
- Open source portable software is now available to general scientific community.
  https://ysph.yale.edu/cmips/research/software/measurement-error-and-misclassification-correction/rrc/